

Modeling of Supersonic Radiative Marshak Waves Using Simple Models and Advanced Simulations

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1 Effective mean free path

if the Rossland mean free path is larger than the halraum diameter, we need to correct it to avoid unphysical results. In other words,

- Optically thick (diffusion regime): $\lambda_R \ll R$, no correction needed.
- Optically thin (transport regime): $\lambda_R \gg R$, need to correct λ_R to be $\sim R$.

So, we can define an effective mean free path as below:

$$\lambda_{\text{eff}} = \left(\frac{1}{\lambda_R} + \frac{1}{R} \right)^{-1}. \quad (1)$$

But, this definition has a problem: when $\lambda_R \ll R$, $\lambda_{\text{eff}} \approx \lambda_R$ (which is good), and when $\lambda_R \gg R$, $\lambda_{\text{eff}} \approx R$ (which is also good). However, in the intermediate regime, λ_{eff} can be significantly smaller than both λ_R and R , which is unphysical. To fix this, we can use the following definition:

$$\lambda_{\text{eff}} = \left(\frac{1}{\lambda_R^n} + \frac{1}{R^n} \right)^{-\frac{1}{n}} \rightarrow \min(\lambda_R, R). \quad (2)$$

This way, λ_{eff} smoothly transitions between λ_R and R without becoming unphysically small in the intermediate regime. The choice of n controls the sharpness of the transition; a larger n results in a sharper transition.

Effective mean-free-path correction in the numerical model

To prevent unphysical behavior when the Rosseland mean free path becomes comparable to or larger than the hohlraum radius, an effective mean-free-path correction is implemented in the numerical model. The Rosseland mean free path is evaluated as

$$\lambda_R(T_s, \rho) = g T_s^\alpha \rho^{-(\lambda+1)}, \quad (3)$$

and the geometric transport scale is taken as the hohlraum diameter,

$$\lambda_{\text{geom}} = 2R. \quad (4)$$

A smooth minimum between these two scales is then defined using a generalized power mean,

$$\lambda_{\text{eff}} = (\lambda_{\text{geom}}^{-p} + \lambda_{\text{R}}^{-p})^{-1/p}, \quad (5)$$

where the numerical parameter p controls the sharpness of the transition. This construction ensures that

$$\lambda_{\text{eff}} \rightarrow \begin{cases} \lambda_{\text{R}}, & \lambda_{\text{R}} \ll 2R, \\ 2R, & \lambda_{\text{R}} \gg 2R, \end{cases}$$

without introducing an artificial minimum at intermediate optical depth.

The opacity prefactor is renormalized consistently with the effective mean free path,

$$g_{\text{eff}}(t) = g \frac{\lambda_{\text{eff}}(t)}{\lambda_{\text{R}}(t)}, \quad (6)$$

so that the diffusion coefficient scales as $D \propto \lambda_{\text{eff}}$ in both the diffusion and geometry-limited regimes.

When enabled (`lam_eff=True`), this correction is applied dynamically at each time step using the instantaneous surface temperature $T_s(t)$ (or $T_s(t - \Delta t)$ in the Marshak marcher). In the ablation case with density evolution, the correction is coupled directly to the density-dependent transport scaling by integrating the combined factor

$$\left(\frac{g_{\text{eff}}}{g} \right) \rho^{\mu - \lambda - 2}$$

inside the time integral used to construct the similarity coefficient, ensuring that geometry-limited transport and density evolution are treated consistently rather than as separable corrections.

Finally, the dimensionless similarity parameter controlling the Marshak-wave front evolution is evaluated as

$$Z_1 = Z_1(C_{\text{eff}}, \rho), \quad (7)$$

so that the wave-front dynamics, opacity, geometry, and density are consistently coupled through λ_{eff} .

Physical meaning: This construction enforces a physically consistent transition between diffusion-dominated transport and free-streaming transport, while preserving correct scaling of opacity, diffusion coefficient, and Marshak-wave similarity structure in cylindrical hohlraum geometry.

2 Results

<i>Diameter</i>	L	$\frac{L}{\lambda_R}$	$\frac{\lambda_R}{D}$	ρ	λ_R [cm]	n	Material	Expt.
0.06	0.04	0.11	5.93	0.05	0.356	$\rightarrow \infty$	C ₆ H ₁₂	Xu
0.06	0.04	0.41	1.633	0.05	0.098	> 2	C ₆ H ₁₂ Cu _{0.394}	Xu
0.16	0.15	8.33	0.1125	0.04	0.018	≥ 2	Ta ₂ O ₅	Back
0.16	0.15	0.83	1.125	0.05	0.18	1.5	SiO ₂	Back
0.30	0.15	1.36	0.3667	0.01	0.11	≥ 2	SiO ₂ low energy	Back
0.20	0.28	1.32	1.06	0.1249	0.212	1	SiO ₂	Moore
0.20	0.28	0.28	4.94	0.1139	0.988	2-2.5	C ₈ H ₇ Cl	Moore
0.08	0.12	0.31	4.875	0.0065	0.39	1	C ₁₅ H ₂₀ O ₆	Keiter
0.08	0.12	2.11	0.7125	0.0625	0.057	2	C ₁₅ H ₂₀ O ₆ Au _{0.172}	Keiter
0.02	0.03	0.64	2.35	0.16	0.047	? ($n > 3$)	C ₈ H ₈	Ji-Yan
0.10	0.12	0.29	4.1	0.029	0.41	1	French gold	French3
0.20	0.2	0.23	4.33	0.0189	0.86	1	French copper	French3

Table 1: Parameters extracted from maximum temperature mean free path calculation.

<i>Diameter</i>	L	$\frac{L}{\lambda_R}$	$\frac{\lambda_R}{D}$	ρ	λ_R [cm]	n	Material	Expt.
0.06	0.04	0.381	1.75	0.05	0.105	$\rightarrow \infty$	C ₆ H ₁₂	Xu
0.06	0.04	1.33	0.50	0.05	0.03	> 2	C ₆ H ₁₂ Cu _{0.394}	Xu
0.16	0.15	10.0	0.0938	0.04	0.015	≥ 2	Ta ₂ O ₅	Back
0.16	0.15	1.88	0.50	0.05	0.08	1.5	SiO ₂	Back
0.30	0.15	3.00	0.1667	0.01	0.05	$\rightarrow \infty$	SiO ₂ low energy	Back
0.20	0.28	1.32	1.06	0.1249	0.212	1	SiO ₂	Moore
0.20	0.28	1.22	1.15	0.1139	0.23	2-2.5	C ₈ H ₇ Cl	Moore
0.08	0.12	1.46	1.025	0.0065	0.082	1	C ₁₅ H ₂₀ O ₆	Keiter
0.08	0.12	4.00	0.375	0.0625	0.03	2	C ₁₅ H ₂₀ O ₆ Au _{0.172}	Keiter
0.02	0.03	2.00	0.75	0.16	0.015	? ($n > 3$)	C ₈ H ₈	Ji-Yan
0.10	0.12	1.14	1.05	0.029	0.105	1	French gold	French3
0.20	0.2	1.33	0.75	0.0189	0.15	1	French copper	French3

Table 2: Parameters extracted from optical depth calculation